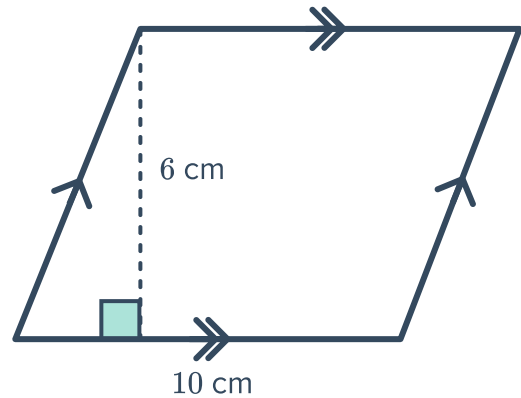


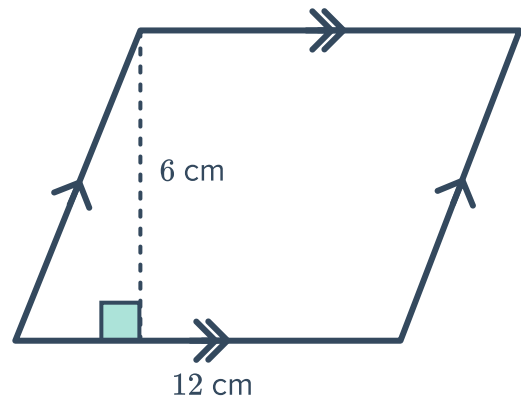
Parallelograms and rectangles



- 1 For the following parallelogram:
 - a If the parallelogram is formed into a rectangle, find:
 - i The length of the rectangle.
 - ii The width of the rectangle.
 - b Find the area of the parallelogram.



- 2 Find the area of the following parallelogram by rearranging it into a rectangle:

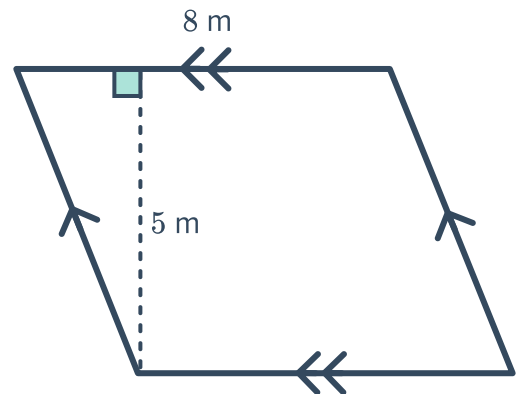




Area of a parallelogram

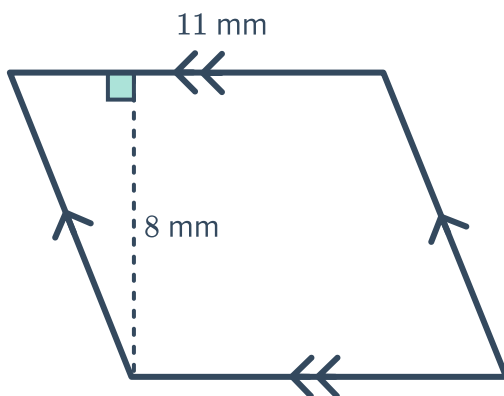
3 Consider the following parallelogram:

- a Identify the base and the height.
- b Find the area of the given parallelogram.

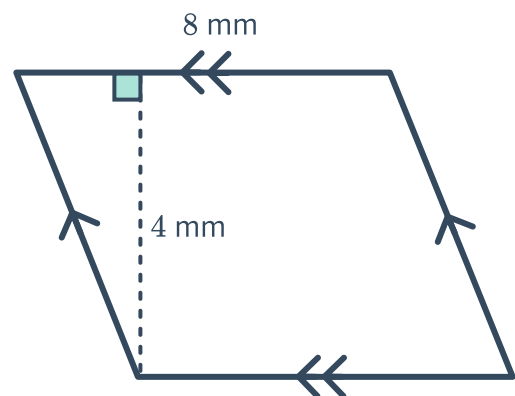


4 Find the area of the following parallelograms:

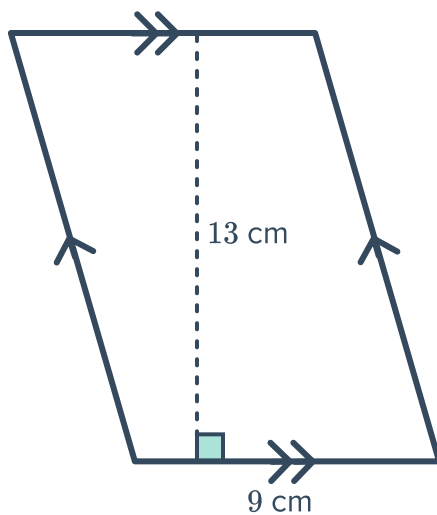
a



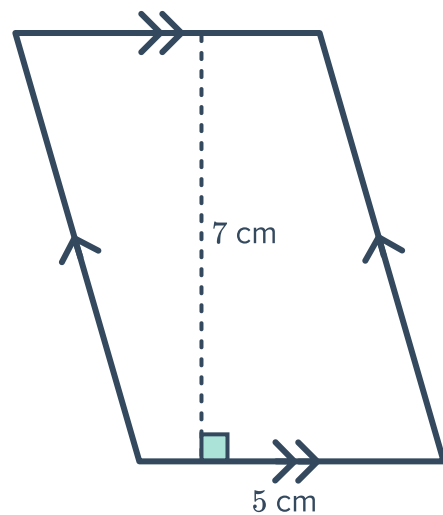
b



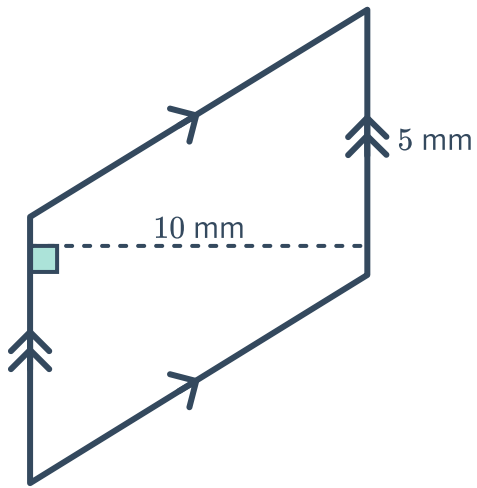
c



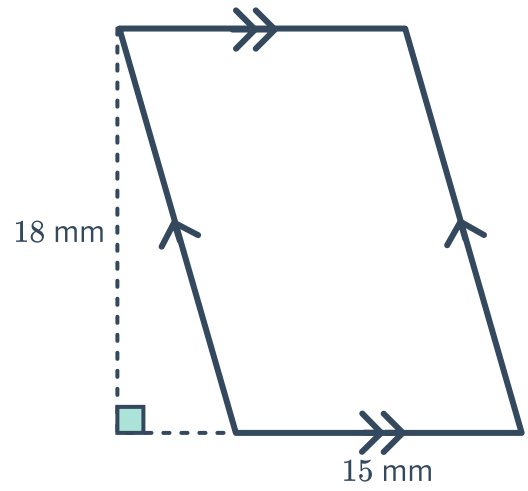
d



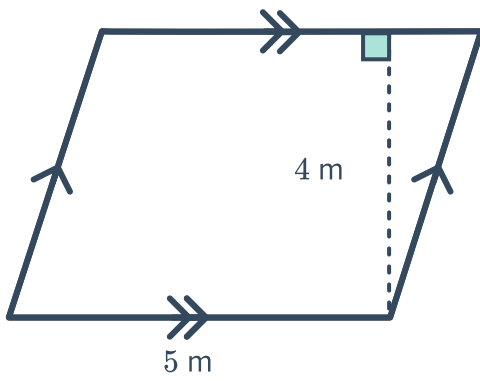
e



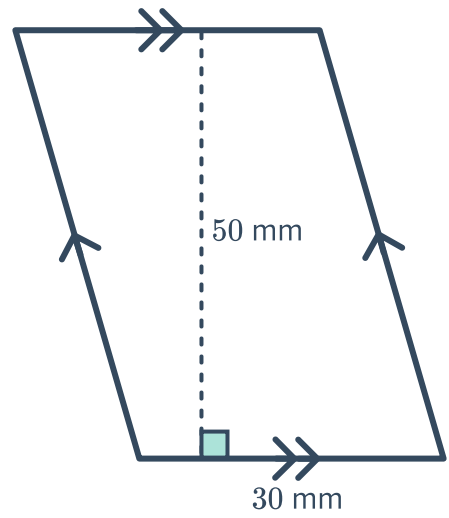
f



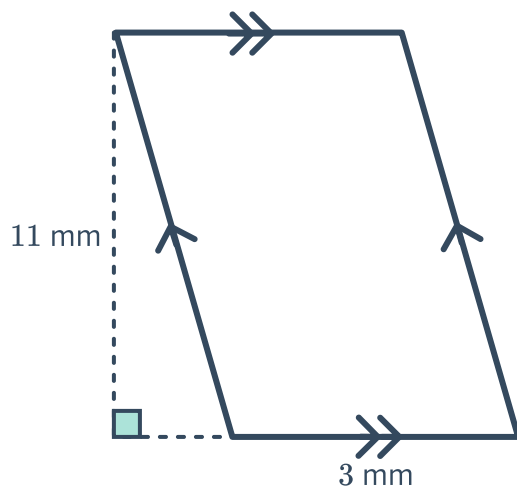
g



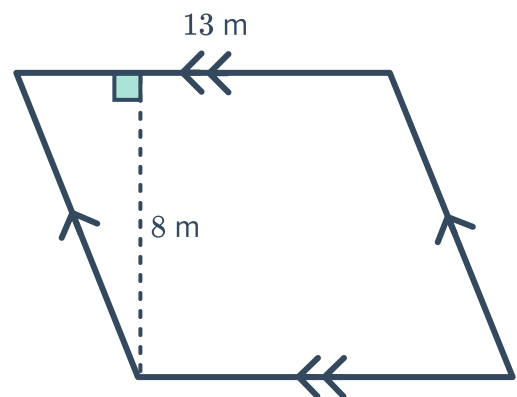
h



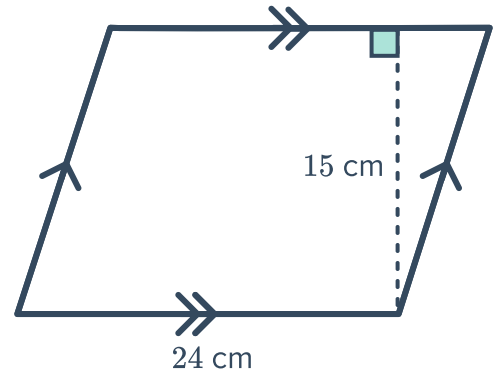
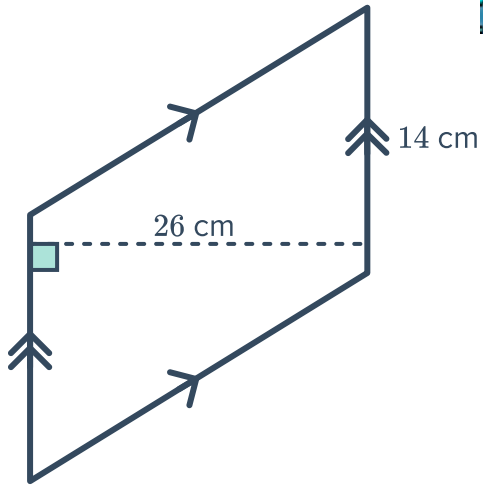
i



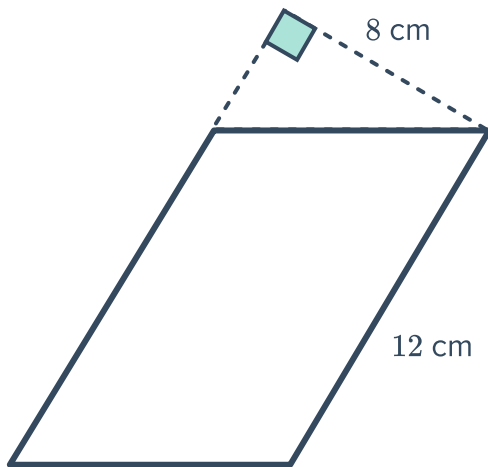
j



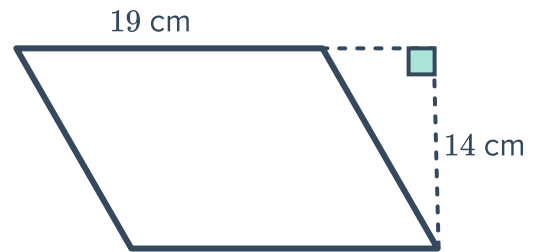
k



m



n



5 Find the area of the parallelograms with the following dimensions:

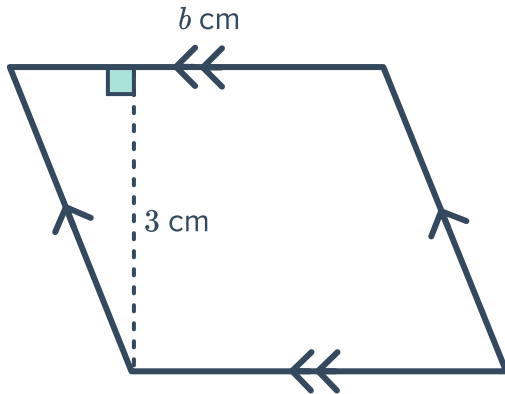
- a The base is 5 mm, and the perpendicular height is 2 mm.
- b The base is 15 cm and the perpendicular height is 7 cm.



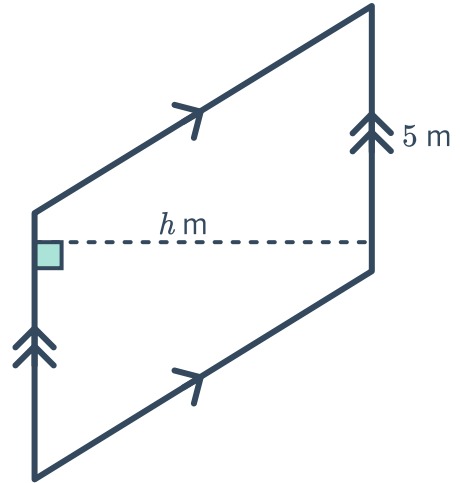
Dimensions from area

6 Find the value of b or h in the following parallelograms:

a Area = 36 cm^2



b Area = 35 m^2



7 Find the base length of a parallelogram whose area is 108 cm^2 and perpendicular height is 9 cm.

8 Find the perpendicular height of a parallelogram whose area is 99 m^2 and base is 9 m.

9 Determine whether the following could be the dimensions of a parallelogram with an area of 28 mm^2 :

a Base = 1 mm, height = 28 mm

b Base = 7 mm, height = 4 mm

c Base = 4 mm, height = 7 mm

d Base = 2 mm, height = 28 mm

10 Complete the following table of base and height measurements for different parallelograms with an area of 60 cm^2 :

Base (cm)		10	30
Height (cm)	12		
Area (cm^2)	60	60	60

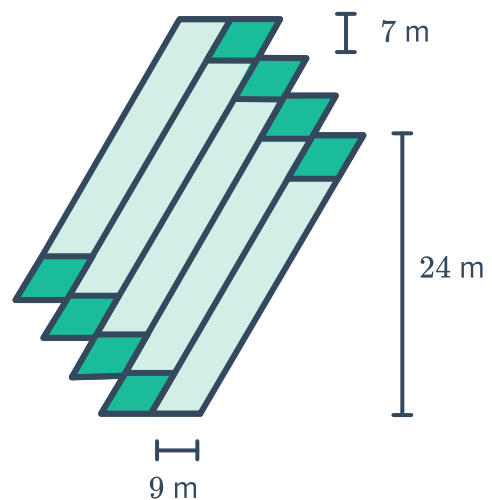


Applications

- 11** Some car parks require the cars to park at an angle as shown. The dimensions of the car park are as given, where each individual parking space has a length of 5.1 m and a width of 3.9 m. What area is needed to create an angled car park suitable for 7 cars?



- 12** The tile pattern shown is made up of two different parallelograms in a tessellated pattern.
- a** Find the area covered by the smaller tiles.
 - b** Find the area covered by the larger tiles.
 - c** Calculate the total area covered.

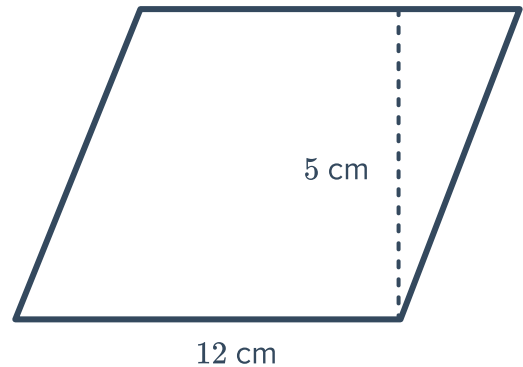


- 13** James used some scrap paper to make a birthday card in the shape of a parallelogram. The base of the card is 17 cm long, and the perpendicular height is 16 cm. Find the area of the card.

- 14 A roof in the shape of a parallelogram is to be entirely covered by identical solar panels that are also in the shape of a parallelogram. Each solar panel measures 1.2 m along the longest side and has a perpendicular height of 0.6 m.
- Find the area covered by one solar panel.
 - Find the number of solar panels to be installed if the roof measures 23.04 m^2 in area, and no part of the roof is to be left uncovered by solar panels.

- 15 An area measuring 2280 cm^2 is to be paved with identical tiles in the shape of parallelograms. Each tile measures 12 cm along the base, and has a perpendicular height of 5 cm.

- Find the area that each tile covers.
- How many tiles are needed to cover the whole area?



- 16 A quilt is made by sewing together 4 identical parallelograms as shown in the following figure:

If the total area of the quilt is 1944 cm^2 , calculate the perpendicular height of each parallelogram piece.

